

Electrical Engineering and Electronics Project

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1. Introduction

The purpose of this project is to design and simulate a regulated DC power supply capable of converting a 220 V AC mains input into a stable +15 V DC output. Such power supplies are widely used in electronic circuits that require a fixed and noise-free DC voltage, including sensors, microcontroller modules, and general laboratory applications.

To achieve this objective, the circuit performs three main processes:

1. Step-down of AC voltage using a 220 V $\rightarrow$ 24 V transformer.
2. AC to DC conversion using a diode bridge rectifier.
3. Voltage regulation and filtering using capacitors, a series resistor, and an LM7815 linear regulator.

The components used in the circuit include (for task 1&2):

- AC input cable
- 220 V / 24 V step-down transformer
- 3 A fuse for safety protection
- Bridge rectifier ($4 \times 1\text{--}2\text{A}$ diodes)
- 1000 μF capacitor
- 100 μF capacitor
- 0.1 μF Capacitor
- 8.1 μF Capacitor
- LM7815 voltage regulator (regulated 15v)
- LM 117 voltage regulator (1.2v-37v)
- 2 k Ω resistor
- 5K ohm potentiometer
- 240 Ohm Resistor
- Heat sink for the regulator

- Breadboard and jumper wires

Together, these components ensure that the final output is a safe, smooth, and constant 15 V DC, suitable for powering low-current DC loads.

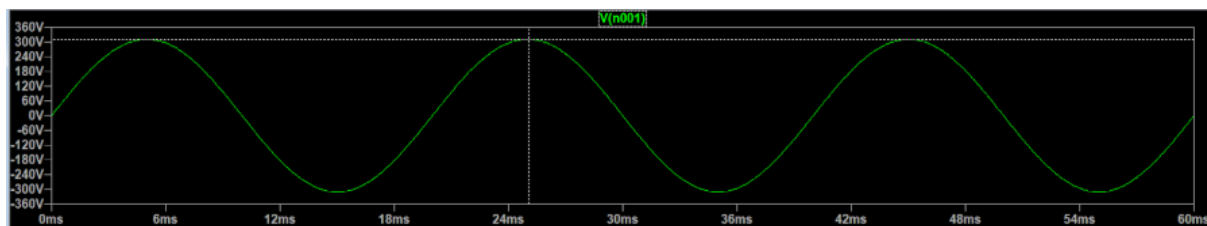
2. LTSpice Simulation and Waveform Explanation

The circuit was simulated on LTSpice to verify each stage of the DC power supply. The waveforms obtained confirm the expected behaviour of every component.

Explanation of the simulation results and the waveforms:

A. AC Input Stage (220 V AC)

Waveform: A sinusoidal waveform at 220 V RMS.

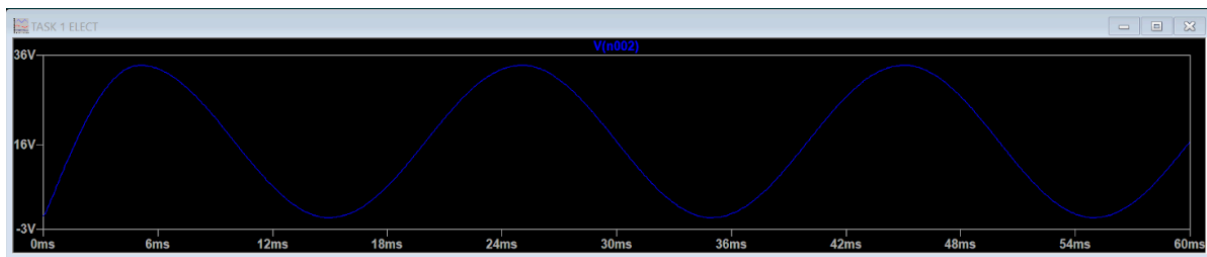


Explanation:

This represents the mains AC supply before any conversion. The waveform is a pure sine wave. At this stage, the fuse protects the circuit from overcurrent conditions but does not affect the waveform unless blown.

B. Step-Down Transformer Output (24 V AC)

Waveform: A sinusoidal waveform at 24 V RMS.



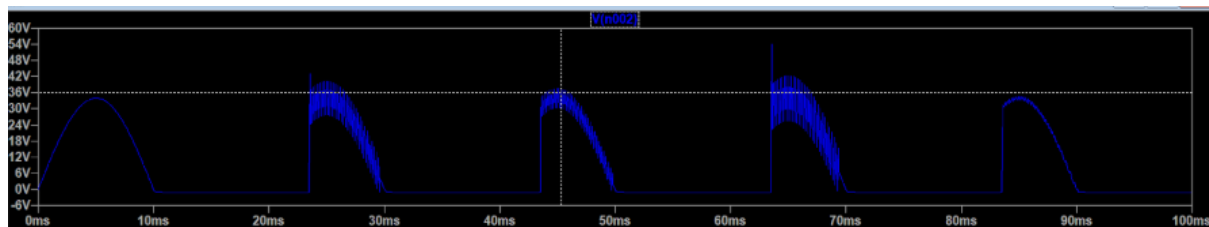
Explanation:

The transformer reduces the mains voltage from 220 V AC to 24 V AC.

Although still sinusoidal and still AC, this lower voltage is now suitable for rectification. The transformer also provides electrical isolation from the mains for safety.

C. Bridge Rectifier Output (Full-Wave Rectified Signal)

Waveform: A full-wave rectified signal – positive “humps” with no negative cycles.



Explanation:

In our circuit, four diodes are connected together to form a full-wave bridge rectifier. This configuration converts the 24 V AC from the transformer into a pulsating DC voltage.

The four diodes in the bridge force both halves of the AC signal to become positive.

The **4 diodes** are connected in a diamond-shaped structure:

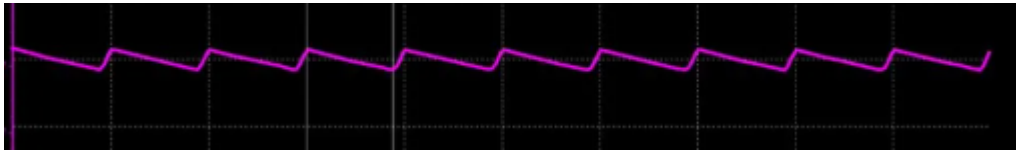
- **Two diodes conduct during the positive half-cycle**
- **The other two conduct during the negative half-cycle**

This ensures that **current always flows in the same direction** toward the load, converting AC into DC.

At this point, the voltage is DC but very unstable and cannot be used to power electronic components, so we will need to use a capacitor.

D. Smoothing capacitor (1000 μ F)

Waveform: A DC level with small ripple.



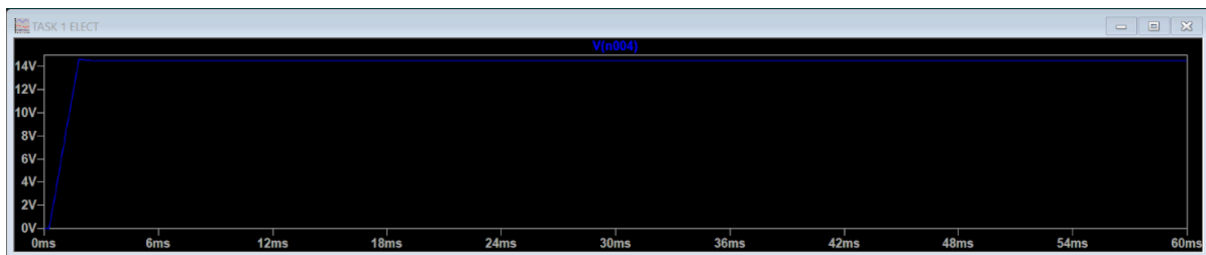
Explanation:

The 1000 μF capacitor charges up to the peak value of the rectified voltage (approximately 34 V) and then releases its stored energy when the input waveform drops between peaks. This charging and discharging action significantly reduces the voltage variations, producing a much smoother DC output.

Although some ripple remains as seen in the waveform, this is because the capacitor cannot maintain a perfectly constant voltage between each charging cycle, but the ripple level is greatly minimized.

E. Output of LM7815 (Regulated +15 V DC)

Waveform: A flat, stable line at +15 V with almost no ripple.



Explanation:

The LM7815 linear regulator removes the remaining ripple and stabilizes the voltage to exactly +15 V.

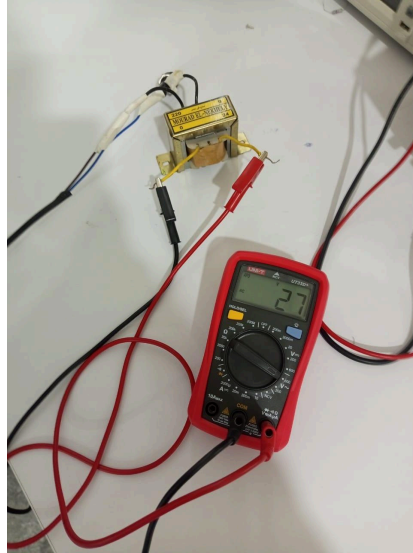
The additional 100 μF capacitor at the output further smooths the signal and improves transient response.

This final result is a clean and constant **15 V DC supply**, which is the main objective of the project.

Summary of the waveforms:

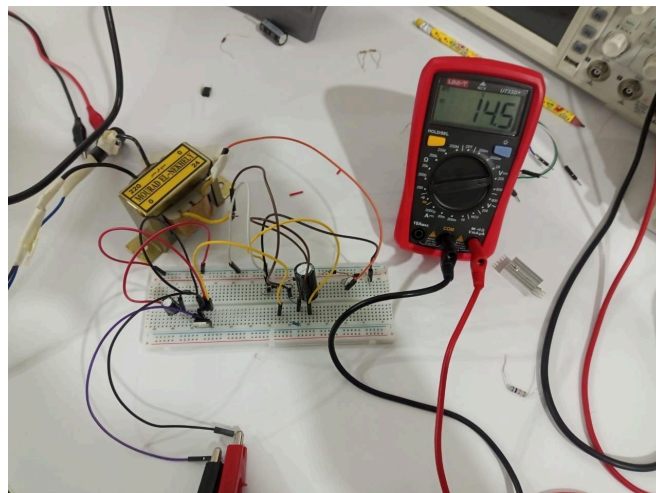
	shape of waves	Polarity of waves	Amplitude
AC Input (220 V AC)	Pure sine wave	Positive and negative cycles	High amplitude(300 v)
Transformer Output (24 V AC)	Same sine wave shape	Positive and negative cycles	Lower amplitude (36 v)
Bridge Rectifier Output (Full-Wave Rectified)	Looks like repeated “humps”	All cycles become positive	The same as 36 v
After Smoothing Capacitor	Mostly DC with small ripple on the top	In the positive side	The same as 36 v
Regulated Output from LM7815 (+15 V DC)	Flat line with no ripple	In the positive side	Very stable 15 V DC
Variable Regulator Output (LM317)	Flat line with no ripple	In the positive side	Level changes when R1&R2 are adjusted

3. Practical - Task 1 and Task 2 (images)



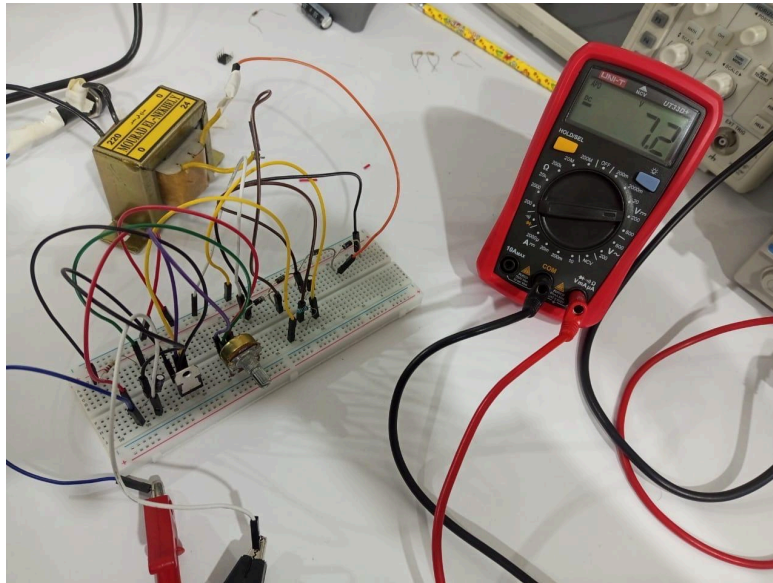
A display of the transformer voltages. The primary is supplied with **220 V RMS**, while the secondary provides approximately **27 V RMS**, which is slightly higher than the rated **24 V** due to no-load conditions.

Task 1:



Circuit diagram of the DC power supply used in Task (1), consisting of a step-down transformer, bridge rectifier, filter capacitors, and an LM7815 voltage regulator to convert **220 V AC** into **15 V DC**.

Task 2:



Circuit diagram of the variable DC power supply used in Task (2), employing a step-down transformer, bridge rectifier, smoothing capacitors, an **LM117 voltage regulator**, and a **5 kΩ potentiometer** to convert **220 V AC** into an adjustable **3–30 V DC** output.

[Task 2 video](#)

Demonstration of the variable DC power supply showing output voltage adjustment using the **5 kΩ potentiometer** and **LM117**. During the demonstration, the output voltage was varied up to **9.2 V DC**, which was confirmed as acceptable by the teaching assistant.

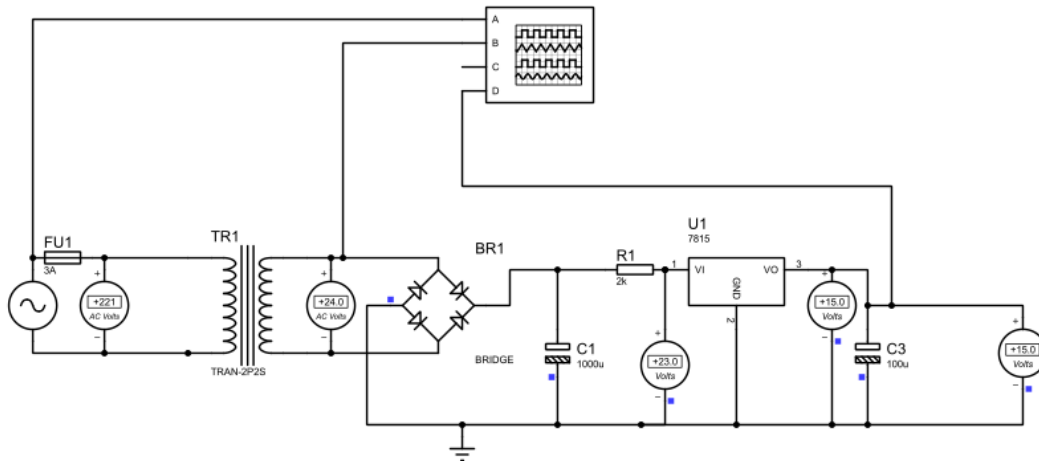
4. Conclusion

Practically implemented to convert a 220 V AC input into a stable +15 V DC output. The simulation performed using LTSpice successfully demonstrated the expected behavior of each stage of the circuit, including the AC input waveform, the step-down transformer output, the full-wave rectifier signal, the smoothing capacitor effect, and the final regulated DC output.

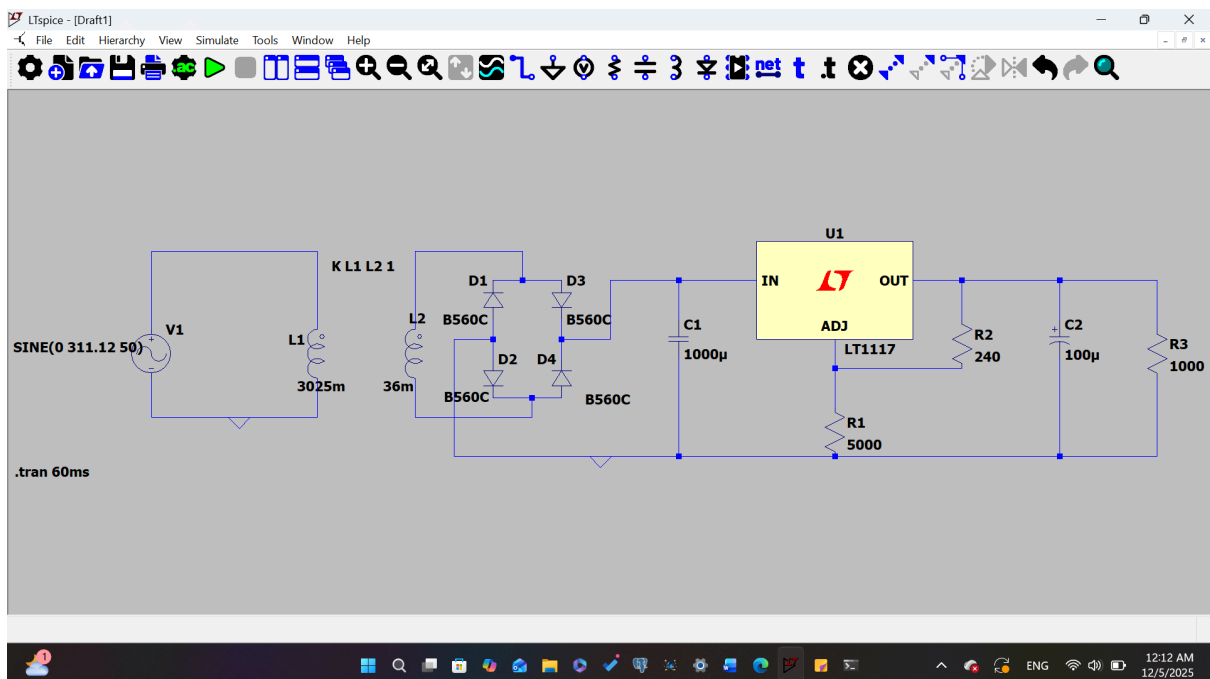
When the circuit was built on the breadboard, the practical oscilloscope readings closely matched the simulation results. The AC signals appeared correctly as sine waves, the rectifier produced the expected full-wave waveform, the filter capacitor significantly reduced ripple, and the LM7815 regulator delivered a clean and constant +15 V DC output. The variable regulator using LM317 also operated correctly, providing a stable and adjustable DC voltage range.

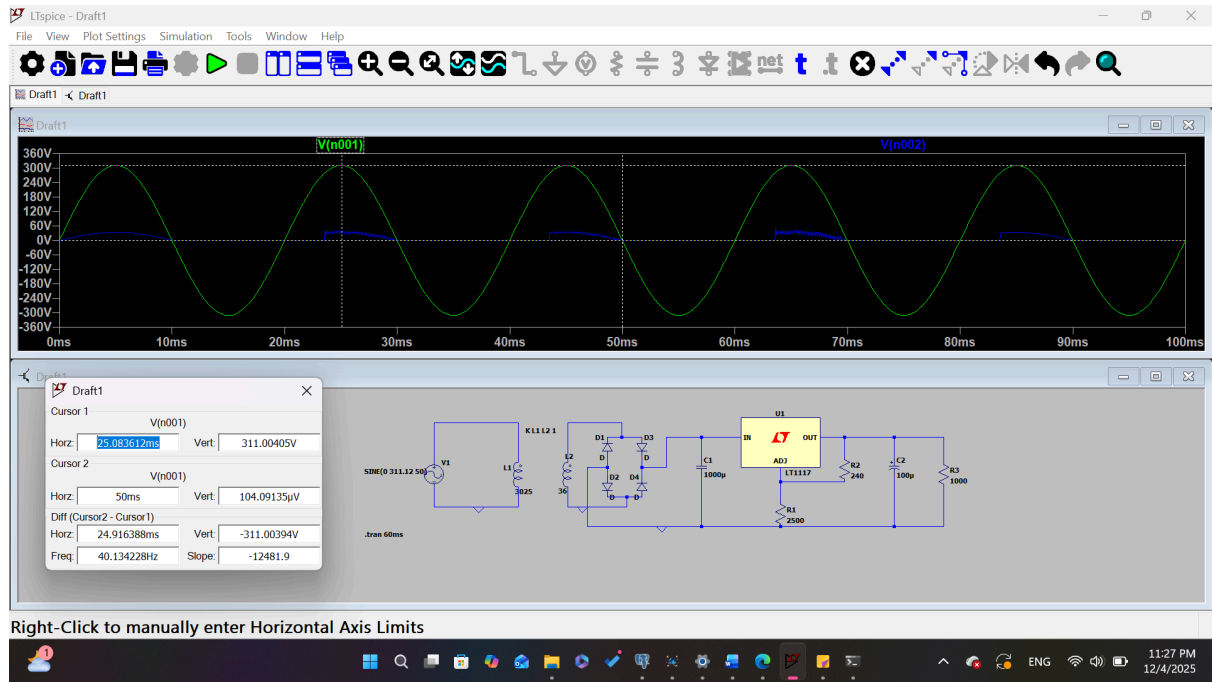
Overall, the project was successful. Both the simulation and practical results confirmed that the design worked as intended. The output waveforms were accurate, stable, and consistent with theoretical expectations. The experiment demonstrated proper AC-to-DC conversion, effective filtering, and reliable voltage regulation.

Task 1

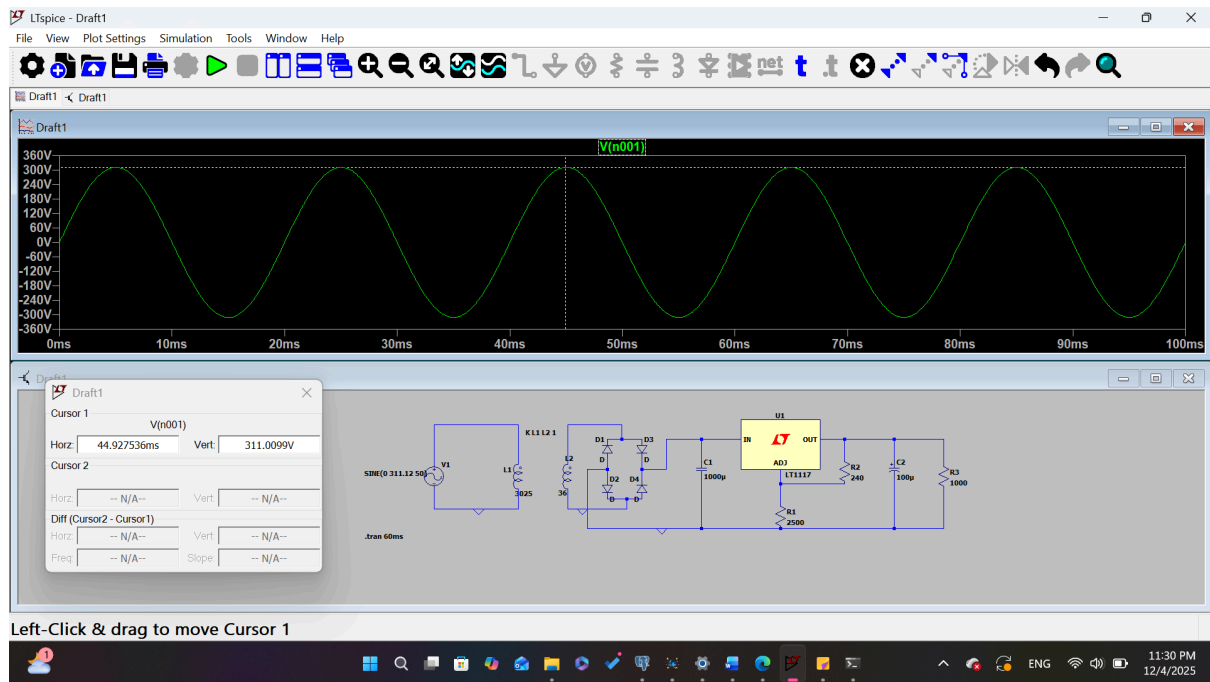


Task 2





Right-Click to manually enter Horizontal Axis Limits



Left-Click & drag to move Cursor 1

